

ACML101
Physical Chemistry Tutorial I

Instructor
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Q#1: How many moles of carbon dioxide are in 66.0 g of dry ice, which is solid CO₂?

Ans: 1.5 mol

Q#2: A student needs 0.366 mol of zinc for a reaction. What mass of zinc in grams should the student obtain?

Ans: 23.9 g

Q#3: Determine the number of molecules in 0.0500 mol of hexane, C₆H₁₄.

Ans: 3.01×10^{22}

Q#4: How many boron atoms are there in 2.00 g of boron?

Ans: 1.11×10^{23}

Q#5: Calculate how many CO₂ molecules there are in 88 g gas.

Ans: 1.2×10^{24}

Q#5: Calculate the how many sodium atoms must react completely with water to give 33.6 dm³ hydrogen gas at STP.

Ans: 3 mol, 3*Avogadro number

Q#6: Sodium burns in oxygen according to the following equation: $4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$

483 g sodium is placed in a container with 5.8 mol oxygen gas. Determine which substance acts as limiting agent. Calculate the mass of sodium oxide that can be produced.

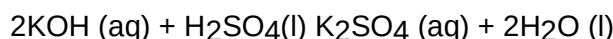
Ans: 10.5 mol, 651 g

Q#7: A standard sodium carbonate solution is prepared by dissolving 5.3 g of the crystals in 250 cm³ water. 25 cm³ of this solution neutralises 10 cm³ of a hydrochloric acid solution.

1. Calculate the concentration of the sodium carbonate solution.
2. Write a balanced equation for the neutralisation reaction.
3. Calculate the concentration of the hydrochloric acid.

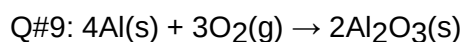
Ans: (1) 0.2 M (3) 1 M

Q#8: 0.28 g of potassium hydroxide is dissolved in water and titrated against a solution of sulphuric acid. The end point is reached after adding exactly 20 cm³ of a sulphuric acid solution. The balanced equation for this reaction is:



Calculate the concentration of the sulphuric acid solution.

Ans: 0.125 M



97.2 g Al reacts and produce 150 g aluminium oxide. Calculate the percentage yield.

Ans: 81.7%

Q#10:

PARAGRAPH V

The reaction of one equivalent of benzene (molar mass = 78 g mol⁻¹) with one equivalent of acetyl chloride in the presence of anhydrous AlCl₃ gave acetophenone (molar mass = 120 g mol⁻¹) as the major product.

Q.30 If the reaction gave 120 g of the acetophenone from 156 g of the benzene, the yield of the reaction is

- (A) 60% (B) 50% (C) 78% (D) 100%

Ans: B

Q#11:

PARAGRAPH III

Titration between oxalic acid dihydrate (a diprotic acid, hereafter referred to as OA) and sodium hydroxide is a typical acid-base titration. Phenolphthalein is used as an indicator. The appearance of a permanent light pink color indicates the end point. This titration method helps in determining the unknown concentration of sodium hydroxide using a standard OA solution. (Molar mass of OA = 126 g mol⁻¹)

~~Q.25~~ The amount (in g) of OA required to make 0.5 L of 0.1 M solution is

- (A) 126 (B) 12.6 (C) 63.0 (D) 6.30

PARAGRAPH III

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Q.26 A titration of 20 mL of 0.1 M solution of OA consumed 16 mL of sodium hydroxide solution. The concentration (in M) of this sodium hydroxide solution is

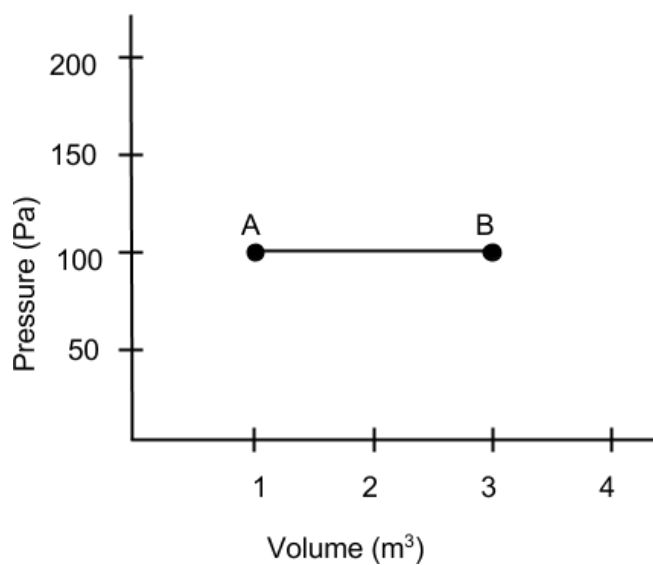
- (A) 0.250 (B) 0.125 (C) 0.0625 (D) 0.160

Ans: (25) 6.3 (26) 0.25

Q#12: In an exothermic process, the volume of a gas expanded from 186 mL to 1997 mL against a constant pressure of 745 torr. During the process, 18.6 calories of heat energy were given off. What was the internal energy change for the system in joules? Also, (1 L atm = 101.3 J)

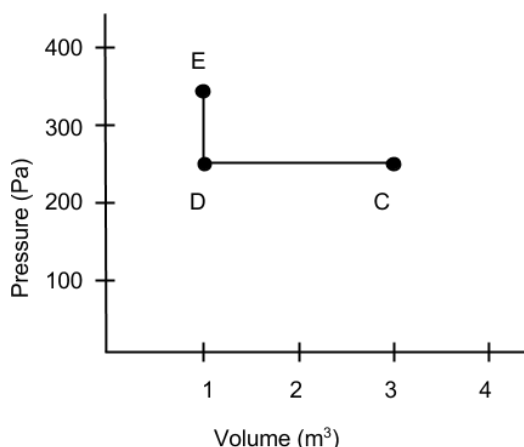
Ans: -257.2 J

Q#13: What is the net work done on the gas as it goes from point A to B on the Pressure vs Volume diagram?



Ans: - 200 J

Q#14: What is the net work done on the gas as it goes from point C to D and then to E on the Pressure vs Volume diagram?



Ans: 500 J

Q#15 For a reversible expansion from 12 litres to 15 litres, what will be the work done by 5 mole of an ideal gas? ($T = 298\text{ K}$)

Ans: -2.76 kJ

Q#16 Determine the difference between enthalpy change and internal energy at constant volume for the combustion of three moles of naphthalene at 298 K.

Ans: -14.86 kJ

Q#17: The standard enthalpy for conversion of Mg to its oxide is -1202 kJ. If $\Delta S^\circ_{\text{sys}}$ for the reaction is -217 J.K^{-1} , comment on the spontaneity of the reaction.

Ans: Spontaneous

Q#18. In the transition from phase α to β , $\Delta G_m(298) = -500\text{ J}$ and $\Delta V_m = 0.05\text{ m}^3$. Calculate the pressure at which a phase becomes more stable form at 298 K.

Ans:

Ans: We know,

$$dG_m = V_m dp - S_m dT$$

At Constant T,

$$dG_m = V_m dp$$

$$dG_{m,\alpha} = V_{m,\alpha} dp \quad (i)$$

$$dG_{m,\beta} = V_{m,\beta} dp \quad (ii)$$

$$dG_{m,\beta} - dG_{m,\alpha} = V_{m,\beta} dp - V_{m,\alpha} dp$$

$$d(G_{m,\beta} - G_{m,\alpha}) = (V_{m,\beta} - V_{m,\alpha}) dp$$

$$d(\Delta G_m) = \Delta V_m dp$$

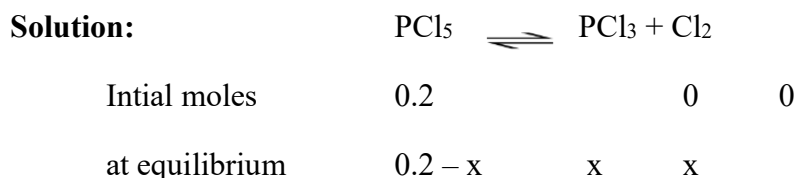
$$\Delta G_{m,P} - \Delta G_{m,P^0} = \Delta V_m (P - P^0)$$

$$P^0 = 1\text{ atm and at } P^0, \Delta G_{m,P} = 0$$

$$(P - P^0) = \frac{-\Delta G_{m,P^0}}{\Delta V_m} = \frac{500}{0.05} = 10^4\text{ Pa} = 0.1\text{ atm}$$

P=1.1 atm

Q#19. Decomposition of 0.2 mol of PCl_5 is taken place in a 10 L flask at 550 K. If pressure of 1 bar is attained at the equilibrium, calculate K_p and K_c for the reaction.



$$\begin{aligned}\text{Total no. of moles at equilibrium} &= (0.2 - x) + x + x \\ &= 0.2 + x\end{aligned}$$

$$\begin{aligned}\text{Also, total no. of moles, } n &= PV/RT \\ &= \frac{1 \text{ bar} \times 10 \text{ L}}{0.083 \text{ bar.L.mol}^{-1} \cdot \text{K}^{-1} \times 550 \text{ K}} = 0.219\end{aligned}$$

$$\therefore 0.2 + x = 0.219$$

$$x = 0.019$$

$$[\text{PCl}_5] = \frac{0.2 - 0.019}{10} \text{ M} = 1.81 \times 10^{-2} \text{ M}$$

$$[\text{PCl}_3] = [\text{Cl}_2] = \frac{0.019}{10} \text{ M} = 1.9 \times 10^{-3} \text{ M}$$

$$K_c = \frac{[\text{PCl}_3][\text{Cl}_2]}{[\text{PCl}_5]} = \frac{(1.9 \times 10^{-3})^2}{1.81 \times 10^{-2}} = 1.99 \times 10^{-4} \text{ M}$$

$$K_p = K_c(RT)^{\Delta n} = 1.99 \times 10^{-4} \times 0.083 \times 550 = 9.1 \times 10^{-3} \text{ bar}$$

Q#20. Decarboxylation of silver carbonate is taking place at 300 K to form silver oxide. The equilibrium constant for the reaction at 300 K is found to be 4.5×10^2 . If the standard enthalpy change for the reaction is $-1.69 \times 10^5 \text{ J.mol}^{-1}$, calculate the temperature at which the equilibrium constant will be 6.93×10^6 .

$$\begin{aligned}\ln \frac{K_2}{K_1} &= -\frac{\Delta H^\circ}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right) \\ \ln \frac{6.93 \times 10^6}{4.5 \times 10^2} &= 9.642 = -\frac{-1.69 \times 10^5}{8.314} \left(\frac{1}{T_2} - \frac{1}{300} \right)\end{aligned}$$

Solving for T_2 , we get

$$T_2 = 262.6 \text{ K}$$

